ECHOES OF TODAY

Members:

Project Manager - Jeric S. Base

Frontend Developer – Jeric S. Base

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Database Manager – Baby Jane Bajaro

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OVERVIEW OF BACKEND SETUP

1. Server Setup (server.js)

- Frameworks & Middleware:

Uses Express.js for the server, CORS for cross-origin requests (allowing requests from http://localhost:5000), and serves static files for the frontend.

- Database Connection:

Connects to MongoDB via a custom connectDB() function (likely in config/db.js).

- Routes:

Mounts user-related API endpoints at /api/users using userRoutes.

- SPA Support:

Serves index.html for any unknown route, supporting client-side routing for a single-page application.

- Port:

Listens on the port defined in .env or defaults to 5000.

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2. Models

a. User.js

Defines the structure for user documents in MongoDB:

- _id: Number (custom, auto-incremented)
- name: String
- email: String
- password: String (hashed)
- role: String

b. Sequence.js

Used for generating auto-incrementing IDs:

- _id: String (e.g., 'userid')
- sequence_value: Number (current value for the sequence)

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3. Routes (userRoutes.js)

Handles all user-related API endpoints:

- Create User (POST /api/users/):
- Increments a sequence to generate a unique numeric user ID.
- Hashes the password with bcrypt.
- Saves the new user to the database.
- Read All Users (GET /api/users/):
- Returns a list of all users.
- Update User (PUT /api/users/:id):
- Updates user data by ID.
- Delete User (DELETE /api/users/:id):
- Deletes a user by ID.
- Login (POST /api/users/login):
- Accepts either username or email and password.
- Finds the user by email or name.
- Compares the provided password with the hashed password.
- Returns user info (excluding password) on success.

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4. Flow Summary

1. Client sends requests to /api/users endpoints.
2. Express server receives and routes requests.
3. Routes handle CRUD and authentication logic, using the User and Sequence models.
4. MongoDB stores user and sequence data.

Code snippets of server.js, sample model, and routes

Server.js

```
// server.js
require('dotenv').config();
const express = require('express');
const cors = require('cors');
const path = require('path');
const connectDB = require('./config/db');

const app = express();

// Connect to MongoDB
connectDB();

// Middleware
app.use(cors({
  origin: 'http://localhost:5000',
  credentials: true
}));
app.use(express.json());

// Routes
const userRoutes = require('./routes/userRoutes');
app.use('/api/users', userRoutes);

// Serve frontend static files
app.use(express.static(path.join(__dirname, '../frontend')));

// Fallback for any unknown routes (for SPA or direct links)
app.get('*', (req, res) => {
  res.sendFile(path.join(__dirname, '../frontend/index.html'));
});

// Start server
const PORT = process.env.PORT || 5000;
app.listen(PORT, () => {
  console.log(`Server running on port ${PORT}`);
});
```

Models

- Sequence.js
- User.js

```
const mongoose = require('mongoose');

const sequenceSchema = new mongoose.Schema({
  _id: String,
  sequence_value: Number
});

module.exports = mongoose.model('Sequence', sequenceSchema);
```

```
const mongoose = require('mongoose');

const userSchema = new mongoose.Schema({
  _id: Number,
  name: String,
  email: String,
  password: String,
  role: String
});

module.exports = mongoose.model('User', userSchema);
```

Routes

- userRoutes.js

```
backend > routes > userRoutes.js > router.post('/') callback
const express = require('express');
const router = express.Router();
const User = require('../models/User');
const Sequence = require('../models/Sequence');
const bcrypt = require('bcrypt');

// Create user
router.post('/', async (req, res) => {
  try {
    const sequenceDocument = await Sequence.findOneAndUpdate(
      { _id: 'userid' },
      { $inc: { sequence_value: 1 } },
      { new: true, upsert: true }
    );

    // Hash the password
    const hashedPassword = await bcrypt.hash(req.body.password, 10);

    const user = new User({
      _id: sequenceDocument.sequence_value,
      name: req.body.name,
      email: req.body.email,
      password: hashedPassword,
      role: req.body.role
    });
    await user.save();
    res.status(201).json(user);
  } catch (err) {
    res.status(400).json({ message: err.message });
  }
});

// Read all users
router.get('/', async (req, res) => {
  const users = await User.find();
  res.json(users);
});

// Update user
router.put('/:id', async (req, res) => {
  try {
    const user = await User.findByIdAndUpdate(req.params.id, req.body, { new: true }
    if (!user) return res.status(404).json({ message: 'User not found' });
    res.json(user);
  } catch (err) {
    res.status(400).json({ message: err.message });
  }
});

// Delete user
router.delete('/:id', async (req, res) => {
  try {
    const user = await User.findByIdAndDelete(req.params.id);
    if (!user) return res.status(404).json({ message: 'User not found' });
```

```
backend > routes > userRoutes.js > ...
router.delete('/:id', async (req, res) => {
    const user = await User.findByIdAndDelete(req.params.id);
    if (!user) return res.status(404).json({ message: 'User not found' });
    res.json({ message: 'User deleted' });
  } catch (err) {
    res.status(400).json({ message: err.message });
  }
});

// Login user
router.post('/login', async (req, res) => {
  const { usernameOrEmail, password } = req.body;

  try {
    console.log('Login attempt:', usernameOrEmail);

    // Try to find by email or username
    const user = await User.findOne({
      $or: [
        { email: usernameOrEmail },
        { name: usernameOrEmail }
      ]
    });

    if (!user) {
      console.log('User not found');
      return res.status(400).json({ message: 'User not found' });
    }

    // Compare passwords
    const isMatch = await bcrypt.compare(password, user.password);
    if (!isMatch) {
      console.log('Invalid credentials');
      return res.status(400).json({ message: 'Invalid credentials' });
    }

    console.log('Login successful');
    res.status(200).json({
      message: 'Login successful',
      user: {
        id: user._id,
        name: user.name,
        email: user.email,
        role: user.role
      }
    });
  } catch (err) {
    console.error('Server error:', err);
    res.status(500).json({ message: 'Server error' });
  }
});

module.exports = router;
```

API Testing Screenshots

```
DELETE http://localhost:5000/api/users/6808fc027c959ea73dc60f95
Status: 200 OK  Size: 26 Bytes  Time: 21 ms

Response
{
  "message": "User deleted"
}
```

```
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
Server running on port 5000
MongoDB Connected
[nodemon] restarting due to changes...
[nodemon] starting `node server.js`
Server running on port 5000
MongoDB Connected
```

File Edit Selection View ··· final-project-Echoes-of-Today

THUNDER CLIENT

New Request

Activity Collections Env

filter activity

GET thunderclient.co... just now

thunderclient.com/welcome × .env U

GET http://localhost:5000/api/users Send

Query Headers 2 Auth Body Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content Format

1

Status: 200 OK Size: 359 Bytes Time: 17 ms

Response Headers 9 Cookies Results Docs

Copy


```
[
  {
    "_id": "6808fa7a7c959ea73dc60f93",
    "name": "Alice",
    "email": "alice@example.com",
    "password": "password123",
    "role": "user",
    "__v": 0
  },
  {
    "_id": "6808fc027c959ea73dc60f95",
    "name": "Jason",
    "email": "jason@gmail.com",
    "password": "lorentepogi",
    "__v": 0
  },
  {
    "_id": "6808fc427c959ea73dc60f97",
    "name": "Jason",
    "email": "jason@gmail.com",
    "password": "jasonpogi123",
    "role": "user",
    "__v": 0
```


Response Chart

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
[nodemon] 3.1.10
[nodemon] to restart at any time, enter `rs`
[nodemon] watching path(s): *.*
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
Server running on port 5000
MongoDB Connected
```

powershell

node

main* ⊗ 0 ⚠ 0 http://127.0.0.1:5500

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GET http://localhost:5000/api/users Send

Query Headers 2 Auth Body Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content Format

1

Status: 200 OK Size: 250 Bytes Time: 10 ms

Response Headers 9 Cookies Results Docs


```
[
  {
    "_id": "6808fa7a7c959ea73dc60f93",
    "name": "Alice",
    "email": "alice@example.com",
    "password": "password123",
    "role": "user",
    "__v": 0
  },
  {
    "_id": "6808fc427c959ea73dc60f97",
    "name": "Jason",
    "email": "jason@gmail.com",
    "password": "jasonpogi123",
    "role": "user",
    "__v": 0
  }
]
```


Response Chart

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
Server running on port 5000
MongoDB Connected
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node

main* ⊗ 0 ⚠ 0 http://127.0.0.1:5500

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THUNDER CLIENT

New Request

Activity Collections Env

filter activity

POST thunderclient.co...
just now

thunderclient.com/welcome .env U

POST http://localhost:5000/api/users Send

Query Headers 2 Auth Body 1 Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content Format

```
{
    "name": "Earl",
    "email": "earl@gmail.com",
    "password": "earlbabaero143",
    "role": "babaero"
}
```

Status: 201 Created Size: 126 Bytes Time: 33 ms

Response Headers 9 Cookies Results Docs

```
{
  "name": "Earl",
  "email": "earl@gmail.com",
  "password": "earlbabaero143",
  "role": "babaero",
  "_id": "6809026afdf6643b558961ae",
  "__v": 0
}
```

Response Chart

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
Server running on port 5000
MongoDB Connected
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Server running on port 5000
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powershell
node

main* 0 0 http://127.0.0.1:5500

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THUNDER CLIENT

New Request

Activity Collections Env

filter activity

GET thunderclient.co...
just now

thunderclient.com/welcome .env U

GET http://localhost:5000/api/users Send

Query Headers 2 Auth Body 1 Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content Format

```
{
    "name": "Earl",
    "email": "earl@gmail.com",
    "password": "earlbabaero143",
    "role": "babaero"
}
```

Status: 200 OK Size: 377 Bytes Time: 13 ms

Response Headers 9 Cookies Results Docs

Copy

```
[
  {
    "_id": "6808fa7a7c959ea73dc60f93",
    "name": "Alice",
    "email": "alice@example.com",
    "password": "password123",
    "role": "user",
    "__v": 0
  },
  {
    "_id": "6808fc427c959ea73dc60f97",
    "name": "Jason",
    "email": "jason@gmail.com",
    "password": "jasonpogi123",
    "role": "user",
    "__v": 0
  },
  {
    "_id": "6809026afdf6643b558961ae",
    "name": "Earl",
    "email": "earl@gmail.com",
    "password": "earlbabaero143",
    "role": "babaero",
```

Response Chart

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

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[nodemon] watching extensions: js,mjs,cjs,json
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MongoDB Connected
```

powershell
node

main* 0 0 http://127.0.0.1:5500

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THUNDER CLIENT

New Request

Activity Collections Env

filter activity

PUT thunderclient.co...
just now

thunderclient.com/welcome .env U

PUT http://localhost:5000/api/users/6808fc027c959ea73dc60f95 Send

Query Headers 2 Auth Body 1 Tests Pre Run

JSON XML Text Form Form-encode GraphQL Binary

JSON Content Format

```
{
  "name": "Michael"
}
```

Status: 200 OK Size: 110 Bytes Time: 25 ms

Response Headers 9 Cookies Results Docs

```
{
  "_id": "6808fc027c959ea73dc60f95",
  "name": "Michael",
  "email": "jason@gmail.com",
  "password": "lorentepogi",
  "__v": 0
}
```

Response Chart

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
[nodemon] watching extensions: js,mjs,cjs,json
[nodemon] starting `node server.js`
Server running on port 5000
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[nodemon] starting `node server.js`
Server running on port 5000
MongoDB Connected
```

powershell
node

main* 0 0 0 http://127.0.0.1:5500

MongoDB Compass - localhost:27017/echoesdb.users

Connections Edit View Collection Help

Compass

My Queries

CONNECTIONS (2)

Search connections

- localhost:27017
  - admin
  - config
  - echoesdb
    - users
  - local
  - yourdbname
- localhost:27017

Welcome users

localhost:27017 > echoesdb > users

Open MongoDB shell

Documents 3 Aggregations Schema Indexes 1 Validation

Type a query: { field: 'value' } or Generate query

Explain Reset Find Options

ADD DATA EXPORT DATA UPDATE DELETE 25 1 - 3 of 3

```
_id: ObjectId('6808fa7a7c959ea73dc60f93')
name : "Alice"
email : "alice@example.com"
password : "password123"
role : "user"
__v : 0
```

```
_id: ObjectId('6808fc427c959ea73dc60f97')
name : "Jason"
email : "jason@gmail.com"
password : "jasonpogi123"
role : "user"
__v : 0
```

```
_id: ObjectId('6809026afdf6643b558961ae')
name : "Earl"
email : "earl@gmail.com"
password : "earlbabaero143"
role : "babaero"
__v : 0
```

GitHub URL to /backend folder :

https://github.com/Chesssster/final-project-Echoes-of-Today/tree/main/backend

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